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Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring

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Abstract

Progressive cerebral infarction (PCI) is a common complication in patients with ischemic stroke that leads to poor prognosis. Blood pressure (BP) can indicate post-stroke hemodynamic changes which play a key role in the development of PCI. The authors aim to investigate the association between BP-derived hemodynamic parameters and PCI. Clinical data and BP recordings were collected from 80 patients with cerebral infarction, including 40 patients with PCI and 40 patients with non-progressive cerebral infarction (NPCI). Hemodynamic parameters were calculated from the BP recordings of the first 7 days after admission, including systolic and diastolic BP, mean arterial pressure, and pulse pressure (PP), with the mean values of each group calculated and compared between daytime and nighttime, and between different days. Hemodynamic parameters and circadian BP rhythm patterns were compared between PCI and NPCI groups using t-test or non-parametric equivalent for continuous variables, Chi-squared test or Fisher's exact test for categorical variables, Cox proportional hazards regression analysis and binary logistic regression analysis for potential risk factors. In PCI and NPCI groups, significant decrease of daytime systolic BP appeared on the second and sixth days, respectively. Systolic BP and fibrinogen at admission, daytime systolic BP of the first day, nighttime systolic BP of the third day, PP, and the ratio of abnormal BP circadian rhythms were all higher in the PCI group. PCI and NPCI groups were significantly different in BP circadian rhythm pattern. PCI is associated with higher systolic BP, PP and more abnormal circadian rhythms of BP.

Original language English

Pages (from-to) 122-133

Number of pages 12

Journal [The Journal of Clinical Hypertension](#)

Volume 26

Issue number 2

Early online date 8 Jan 2024

DOIs • <https://doi.org/10.1111/jch.14759>

Publication status E-pub ahead of print - 8 Jan 2024

Bibliographical note

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Funder

The authors would like to thank the radiology department for their help. This research was funded by Medical and Health Science and Technology Plan Project of Zhoushan and Medical Health Science and Technology Project of Zhejiang Provincial Health Commission, grant number 2021YA07, 2022KY1367.

Keywords

- stroke
- hemodynamic parameters
- blood pressure (BP)
- progressive cerebral infarction (PCI)

Access to Document

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Li, L., Wu, B., Dong, J., He, S., Xu, J., Tse, G., Dai, F., & Liu, H. (2024). [Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring](https://doi.org/10.1111/jch.14759). **The Journal of Clinical Hypertension**, 26(2), 122-133. Advance online publication. <https://doi.org/10.1111/jch.14759>
Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring. / Li, Ling; Wu, Bin; Dong, Jiaoxuan et al.
 In: [The Journal of Clinical Hypertension](https://doi.org/10.1111/jch.14759), Vol. 26, No. 2, 08.01.2024, p. 122-133.

Research output: Contribution to journal > Article > peer-review

Li, L, Wu, B, Dong, J, He, S, Xu, J, Tse, G, Dai, F & Liu, H 2024, 'Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring', **The Journal of Clinical Hypertension**, vol. 26, no. 2, pp. 122-133. <https://doi.org/10.1111/jch.14759>
 Li L, Wu B, Dong J, He S, Xu J, Tse G et al. [Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring](https://doi.org/10.1111/jch.14759). [The Journal of Clinical Hypertension](https://doi.org/10.1111/jch.14759). 2024 Jan 8;26(2): 122-133. Epub 2024 Jan 8. doi: 10.1111/jch.14759
 Li, Ling ; Wu, Bin ; Dong, Jiaoxuan et al. / **Hemodynamic changes in progressive cerebral infarction : An observational study based on blood pressure monitoring**. In: [The Journal of Clinical Hypertension](https://doi.org/10.1111/jch.14759). 2024 ; Vol. 26, No. 2. pp. 122-133.
 @article{03fe8bc41c5a46a992b8c8cecbf0ffe4,
 title = "Hemodynamic changes in progressive cerebral infarction: An observational study based on blood pressure monitoring",
 abstract = "Progressive cerebral infarction (PCI) is a common complication in patients with ischemic stroke that leads to poor prognosis. Blood pressure (BP) can indicate post-stroke hemodynamic changes which play a key role in the development of PCI. The authors aim to investigate the association between BP-derived hemodynamic parameters and PCI. Clinical data and BP recordings were collected from 80 patients with cerebral infarction, including 40 patients with PCI and 40 patients with non-progressive cerebral infarction (NPCI). Hemodynamic parameters were calculated from the BP recordings of the first 7 days after admission, including systolic and diastolic BP, mean arterial pressure, and pulse pressure (PP), with the mean values of each group calculated and compared between daytime and nighttime, and between different days. Hemodynamic parameters and circadian BP rhythm patterns were compared between PCI and NPCI groups using t-test or non-parametric equivalent for continuous variables, Chi-squared test or Fisher's exact test for categorical variables, Cox proportional hazards regression analysis and binary logistic regression analysis for potential risk factors. In PCI and NPCI groups, significant decrease of daytime systolic BP appeared on the second and sixth days, respectively. Systolic BP and fibrinogen at admission, daytime systolic BP of the first day, nighttime systolic BP of the third day, PP, and the ratio of

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keywords = "stroke, hemodynamic parameters, blood pressure (BP), progressive cerebral infarction (PCI)",

author = "Ling Li and Bin Wu and Jiaoxuan Dong and Songbin He and Jie Xu and Gary Tse and Fangyu Dai and Haipeng Liu",

note = "{\textcopyright} 2023 The Authors. The Journal of Clinical Hypertension published by Wiley Periodicals LLC. This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.",

year = "2024",

month = jan,

day = "8",

doi = "10.1111/jch.14759",

language = "English",

volume = "26",

pages = "122--133",

journal = "The Journal of Clinical Hypertension",

issn = "1524-6175",

publisher = "Wiley",

number = "2",

}

TY - JOUR

T1 - Hemodynamic changes in progressive cerebral infarction

T2 - An observational study based on blood pressure monitoring

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PY - 2024/1/8

Y1 - 2024/1/8

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KW - stroke

KW - hemodynamic parameters

KW - blood pressure (BP)

KW - progressive cerebral infarction (PCI)

UR - <http://www.scopus.com/inward/record.url?scp=85181724715&partnerID=8YFLogxK>

U2 - 10.1111/jch.14759

DO - 10.1111/jch.14759

M3 - Article

SN - 1524-6175

VL - 26

SP - 122

EP - 133

JO - The Journal of Clinical Hypertension

JF - The Journal of Clinical Hypertension

IS - 2

ER -



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